

Financial Accounting

Executive MBA

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Lecture 9: Analyzing Financial Statements



The University of Chicago Booth School of Business

Topics: Measuring Performance from Financial Statements

- Profitability Analyses:
 - Return on equity (ROE) vs. Return on assets (ROA).
 - Relation between ROA and ROE.
- Liquidity and Solvency Analyses:
 - Financial Leverage
 - Measuring liquidity and solvency from financial statements.

Return on Equity (ROE)

- ROE measures the earnings available to common equity shareholders divided by the book value of the common equity investment (shareholders' equity):

$$\text{ROE} = \frac{\text{Net income available to common equity shareholders}}{\text{Average stockholders' equity}}$$

ROE Computation

- For companies with *non-controlling interest*

$$\text{ROE} = \frac{\text{Net income attributable to company shareholders}}{\text{Average equity attributable to company shareholders}}$$

- For companies with *preferred stock*

$$\text{ROE} = \frac{\text{Net income attributable to company shareholders} - \text{Preferred dividends}}{\text{Average equity attributable to company shareholders} - \text{Average preferred equity}}$$

ROE: Pfizer

Financial Statement Data for Pfizer's ROE Computation

\$ millions	Dec. 29, 2022	Dec. 30, 2021
Revenues	\$100,330	\$ 81,288
Net income before allocation to noncontrolling interest	31,407	22,025
Net income attributable to Pfizer Inc. common shareholders	31,372	21,979
Total assets.	197,205	181,476
Total Pfizer Inc. shareholders' equity	95,661	77,201

$$\text{ROE} = \frac{\$31,372}{(\$95,661 + \$77,201)/2} = 36.3\%$$

ROE for **S&P 500 firms** ranged from 13.7% to 17.3% from 2018-22.

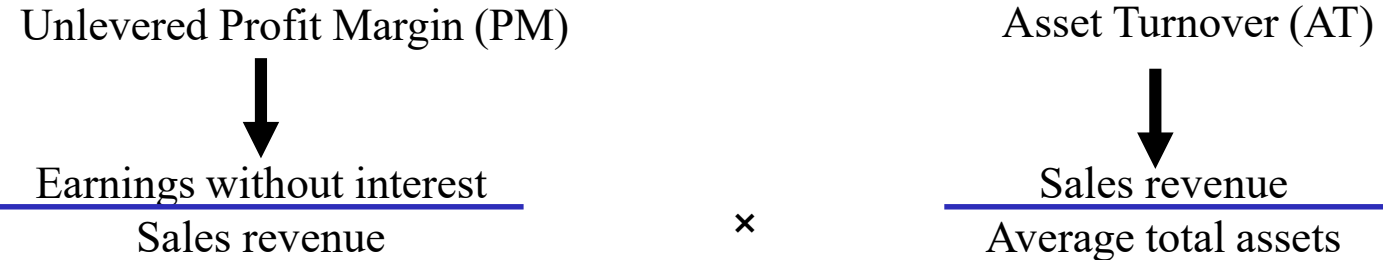
Return on Assets (ROA)

- ROA is equal to net income on all assets (available to all investors) divided by assets:
 - Income should include income attributable to *all* debtholders and non-controlling shareholders because total assets includes minority interests

$$\text{ROA} = \frac{\text{Net income} + [\text{Interest expense} \times (1 - \text{statutory tax rate})]}{\text{Average total assets}}$$

Disaggregating ROA: Traditional Dupont Analysis

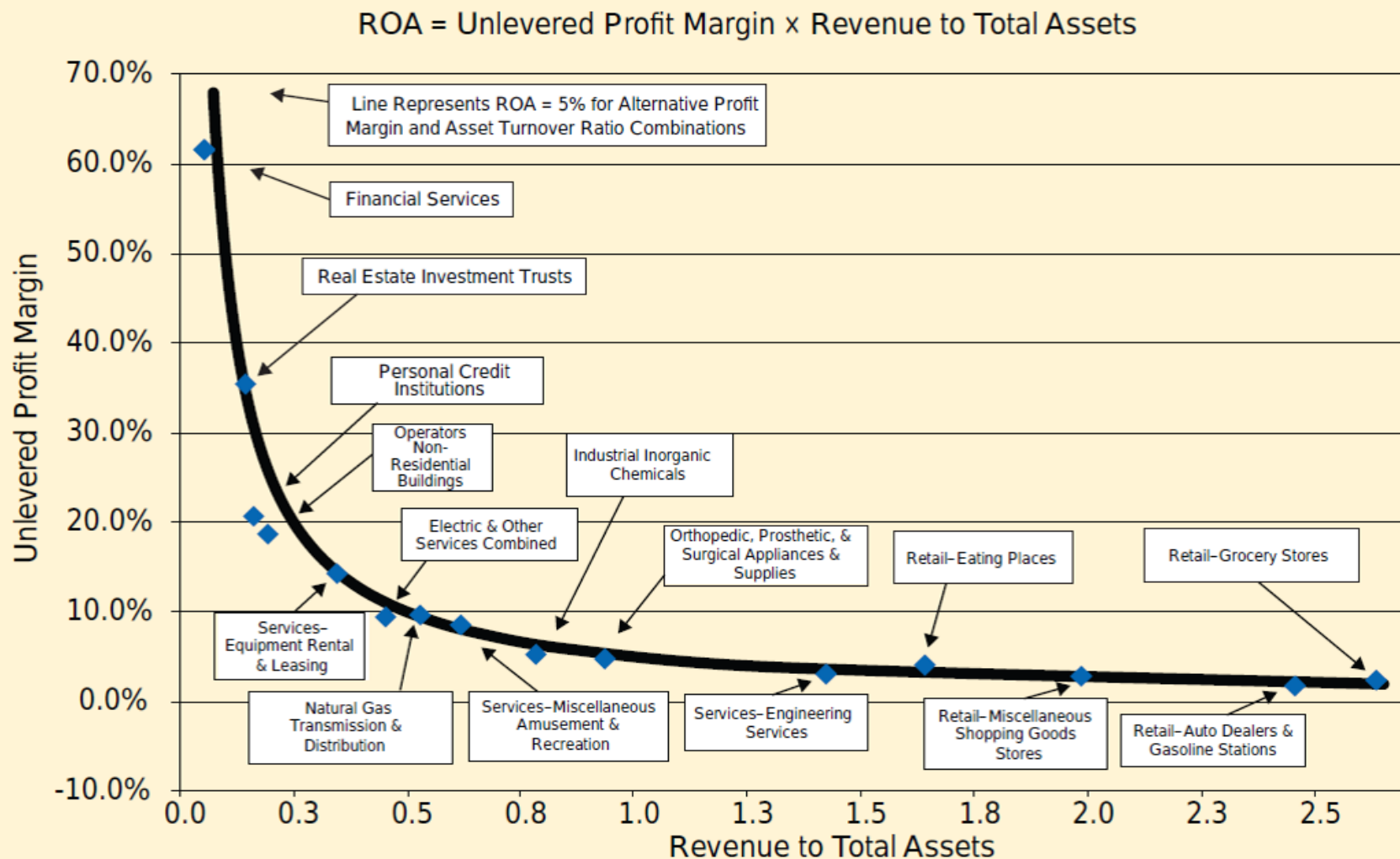
$$\begin{aligned} \text{ROA} &= \frac{\text{Earnings without interest expense}}{\text{Average total assets}} \\ &= \frac{\text{Unlevered Profit Margin (PM)}}{\text{Sales revenue}} \times \frac{\text{Asset Turnover (AT)}}{\text{Average total assets}} \end{aligned}$$



Managers can increase ROA by

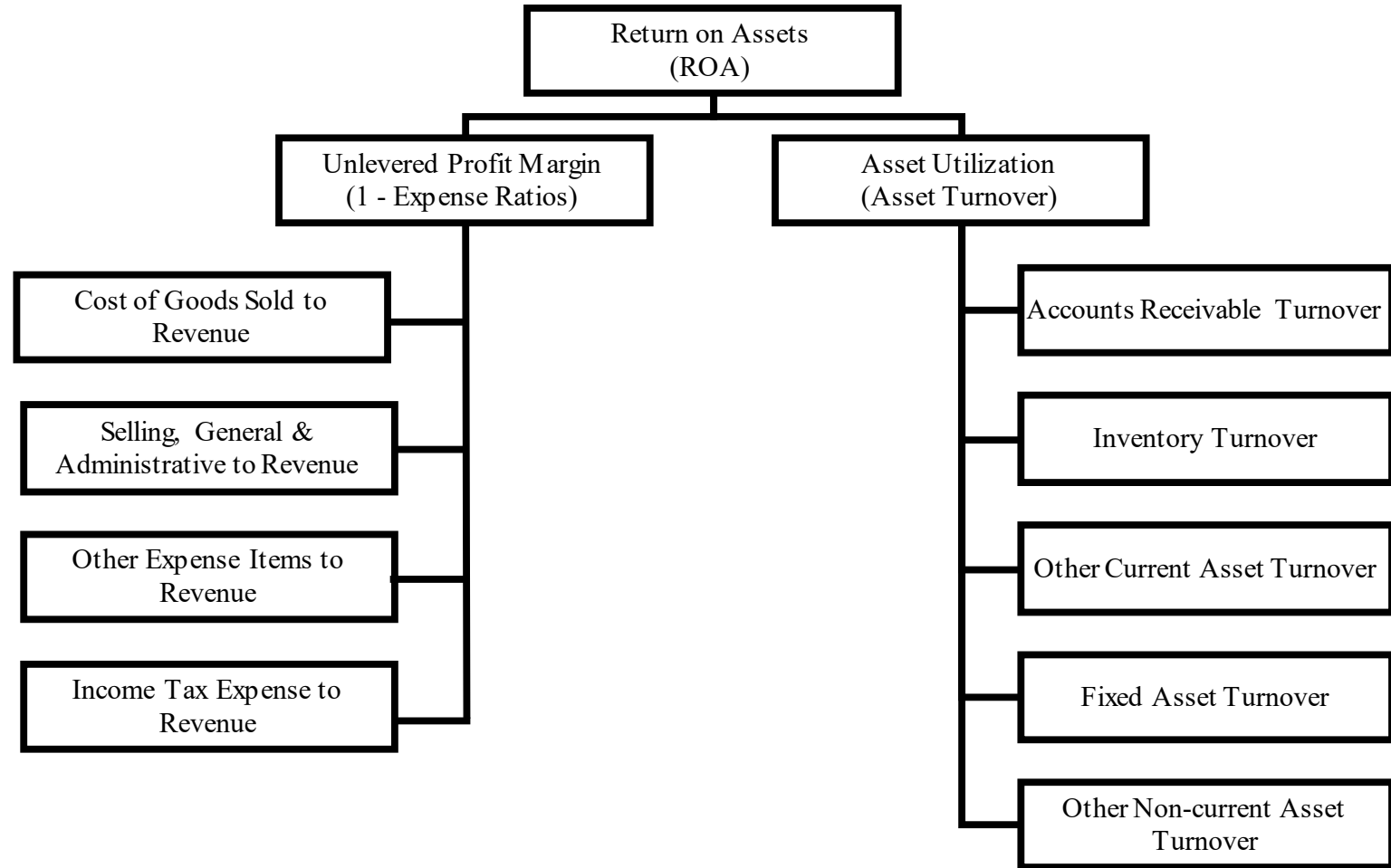
- Increasing PM: increase profitability for a given level of sales
- Increasing AT: reduce assets while still generating same sales level

Tradeoff between Unleveraged Profit Margin vs. Asset Turnover



Source: Holthausen and Zmijewski, *Corporate Valuation: Theory, Evidence & Practice*, 2020.

Disaggregating ROA even further to analyze profitability and productivity drivers...

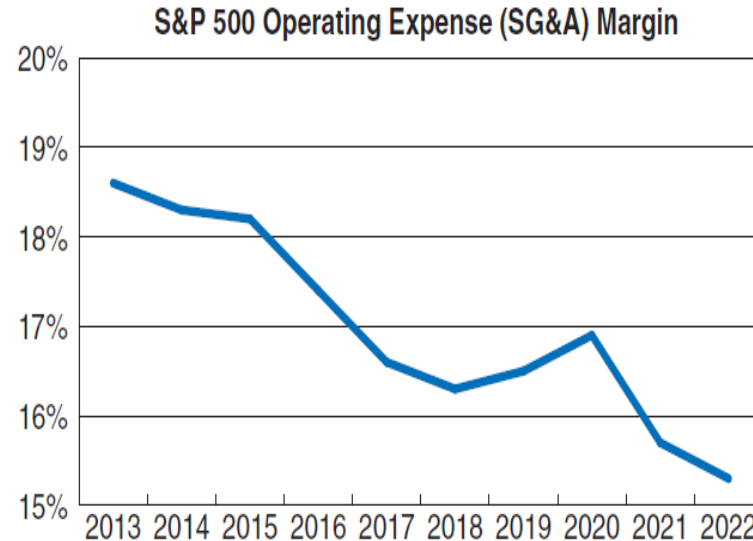


Analysis of Profitability

- Gross Profit Margin = Gross profit / Sales
 - Influenced by *both* the selling price of a company's products and the cost to make or buy those products
 - Generally high and increasing gross profit margin is better
 - Low or decreasing gross profit margin signals more competition or less demand for the company's products
 - Competitive intensity has increased
 - Product line has lost appeal
 - Product costs have increased
 - Product mix has changed
 - Volume has declined and fixed costs have not

Analysis of Profitability...

- Operating Expense Margin = SG&A expense / Sales
 - Measures general operating costs for each sales dollar
 - Consider each expense in whatever detail the company provides in its income statement
 - Compare margins over time and against peers



Analysis of Productivity: Asset Turnover

- Asset Turnover (AT) = $\frac{\text{Sales}}{\text{Average Total Assets}}$
- Turnover = $\frac{\text{Income statement item}}{\text{Average Balance sheet item}}$  **Natural linkage**
- Working capital turnover ratios

Receivables turnover

$$= \frac{\text{Sales}}{\text{Average A/R}}$$

Inventory turnover

$$= \frac{\text{COGS}}{\text{Average Inventory}}$$

A/P turnover

$$= \frac{\text{Purchases}}{\text{Average A/P}}$$

Receivables Turnover (RT) & Days Sales Outstanding (DSO)

$$RT = \frac{\text{Credit Sales revenue}}{\text{Average accounts receivable}}$$

- Receivables turnover (RT) measures how many times receivables have been collected during a period.

$$DSO = \frac{365}{RT}$$

- Days Sales Outstanding (DSO) measures the average number of days it takes to collect receivables during a period which can be a quarter (90 days) or a year (365).

Inventory Turnover (INVT) & Days Inventory Outstanding (DIO)

$$\text{INVT} = \frac{\text{COGS}}{\text{Average inventory}}$$

- Inventory turnover (IT) measures the flow of goods out of inventory during a period relative to the average balance that is held in inventory.

$$\text{DIO} = \frac{365}{\text{INVT}}$$

- Days Inventory Outstanding (DIO) measures the average days it takes to sell inventory during a year.

Accounts Payable Turnover (APT) & Days Payable Outstanding (DPO)

$$\text{APT} = \frac{\text{Purchases}}{\text{Average Accounts payable}}$$

- Accounts payable turnover (APT) measures the number of times purchases are paid during the period.

$$\text{DPO} = \frac{365}{\text{APT}}$$

- Days Payable Outstanding (DPO) measures the average number of days it takes to pay suppliers during a year.

Cash Conversion Cycle (CCC)

CCC is the average number of days to:

- buy inventory on credit (accounts payable)
- sell inventories of credit (accounts receivable)
- collect the receivables
- pay the accounts payable

$$\text{DSO} = \frac{365}{\text{A/R turnover}}$$

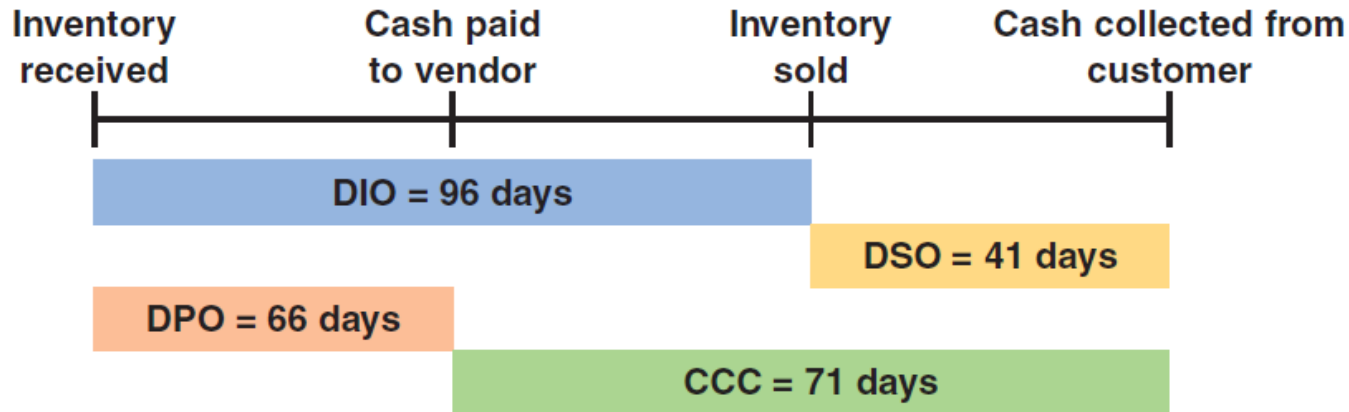
$$\text{DIO} = \frac{365}{\text{Inventory turnover}}$$

$$\text{DPO} = \frac{365}{\text{A/P turnover}}$$

$$\text{CCC} = \text{DSO} + \text{DIO} - \text{DPO}$$

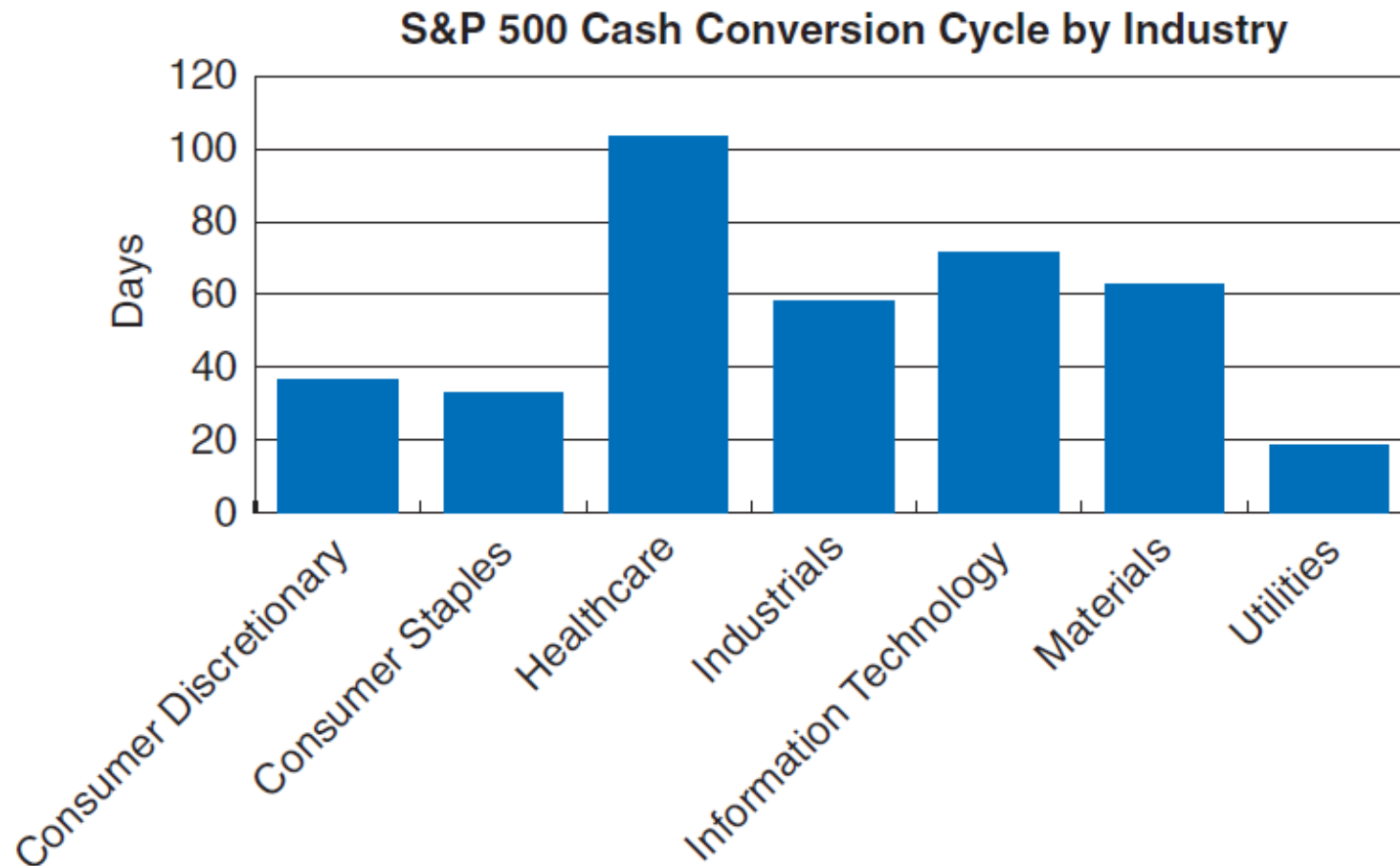
Generally companies *prefer a lower cash conversion cycle* because then the operating cycle is generating profit and cash flow quickly. A useful analysis of this measure focuses on trends over time and comparisons to peers (with similar business models).

Cash Conversion Cycle: Pfizer



- Pfizer holds **inventory** of raw materials, work in process and finished goods for 96 days
- Pfizer's suppliers provide financing for 66 days and customers pay in 41 days
- Pfizer's cash is tied up in inventories and receivables for 71 days which it must finance
- A positive cash conversion cycle is the norm for most firms

Typical Cash Conversion Cycle by industry...



Apple's negative cash conversion cycle for fiscal 2019

Days Sales Outstanding (DSO)	28.2
+ Days Inventory Outstanding (DIO)	9.8
– Days Payables Outstanding (DPO)	(111.6)
= Cash Conversion Cycle	(73.6)

- Apple carries little inventory as its products are pre-sold and shipped when manufactured.
- Consequently, its quick sale of inventory and relatively longer time to pay suppliers results in a negative cash conversion cycle of (73.6) days.
- The negative number means that Apple is able to invest the cash it receives from the sale of its products for 73.6 days on average before that cash is needed to pay suppliers.
- This allows Apple to generate both profit from the sale *and* profit from investing cash.
- A negative cash conversion cycle is generally viewed positively.

What is the relationship between ROA vs. ROE?

- ROE typically *exceeds* ROA because of *financial leverage* (*amount of debt vs. equity used to finance a company's operations*) where the excess return is called the Return on Financial Leverage (ROFL):

$$\text{ROFL} = \text{ROE} - \text{ROA}$$

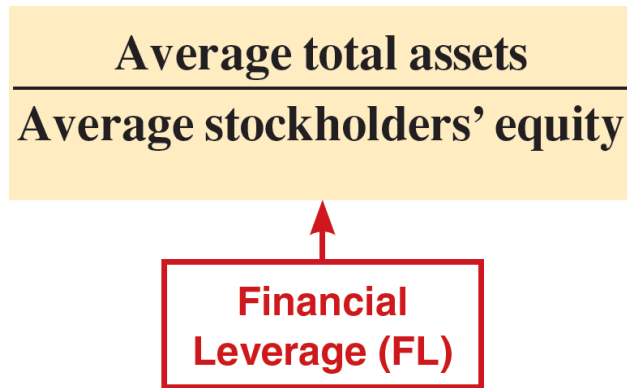
- Financial Leverage *magnifies* the impact of changes on ROA on ROE. When ROA is high, ROE will typically be even higher. But, when ROA is low, ROE will be even lower and perhaps even negative:
 - Therefore $\text{ROFL} > 0$ implies that $\text{ROE} > \text{ROA}$ while $\text{ROFL} < 0$ implies that $\text{ROE} < \text{ROA}$.

Financial Leverage

$$\frac{\text{Average total assets}}{\text{Average stockholders' equity}}$$

↑

**Financial
Leverage (FL)**



- Financial leverage measures the degree to which the company finances its assets with debt versus equity
- Financial leverage is important because debt is a contractual obligation and a company's failure to repay principal or interest can result in legal repercussions
- All else equal, higher financial leverage increases the probability of default and possible bankruptcy

Scenario 1: ROA > Cost of Debt

- Company A is financed with 100% equity. Company B is financed 50% with debt. Tax rate, $t = 20\%$.

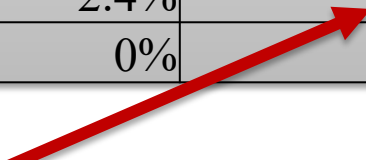
	Company A	Company B
Assets (average)	\$1,000	\$1,000
Net Operating Profits Before Taxes (NOPBT)	\$100	\$100
Equity (average)	\$1,000	\$500
Debt (average)	\$0	\$500
Interest expense I (4% of debt face value)	\$0	\$20
Net income = $(NOPBT - I)(1 - t)$	\$80	\$64
ROA = $[\text{Net Income} + I(1 - t)] / (\text{Assets})$	8.0%	8.0%
ROE = Net income/equity	8.0%	12.8%
ROFL = ROE – ROA	0%	4.8%

Company B made effective use of its financing to improve its ROE by 4.8%.

Scenario 2: ROA < Cost of Debt

- Company A is financed with 100% equity. Company B is financed 50% with debt. Tax rate, $t = 20\%$.

	Company A	Company B
Assets (average)	\$1,000	\$1,000
Net Operating Profits Before Taxes (NOPBT)	\$30	\$30
Equity (average)	\$1,000	\$500
Debt	\$0	\$500
Interest expense I (4% of debt face value)	\$0	\$20
Net income = $(NOPBT - I)(1 - t)$	\$24	\$8
ROA = $[\text{Net Income} + I(1 - t)] / (\text{Assets})$	2.4%	2.4%
ROE = Net income/equity	2.4%	1.6%
ROFL = ROE - ROA	0%	-0.8%



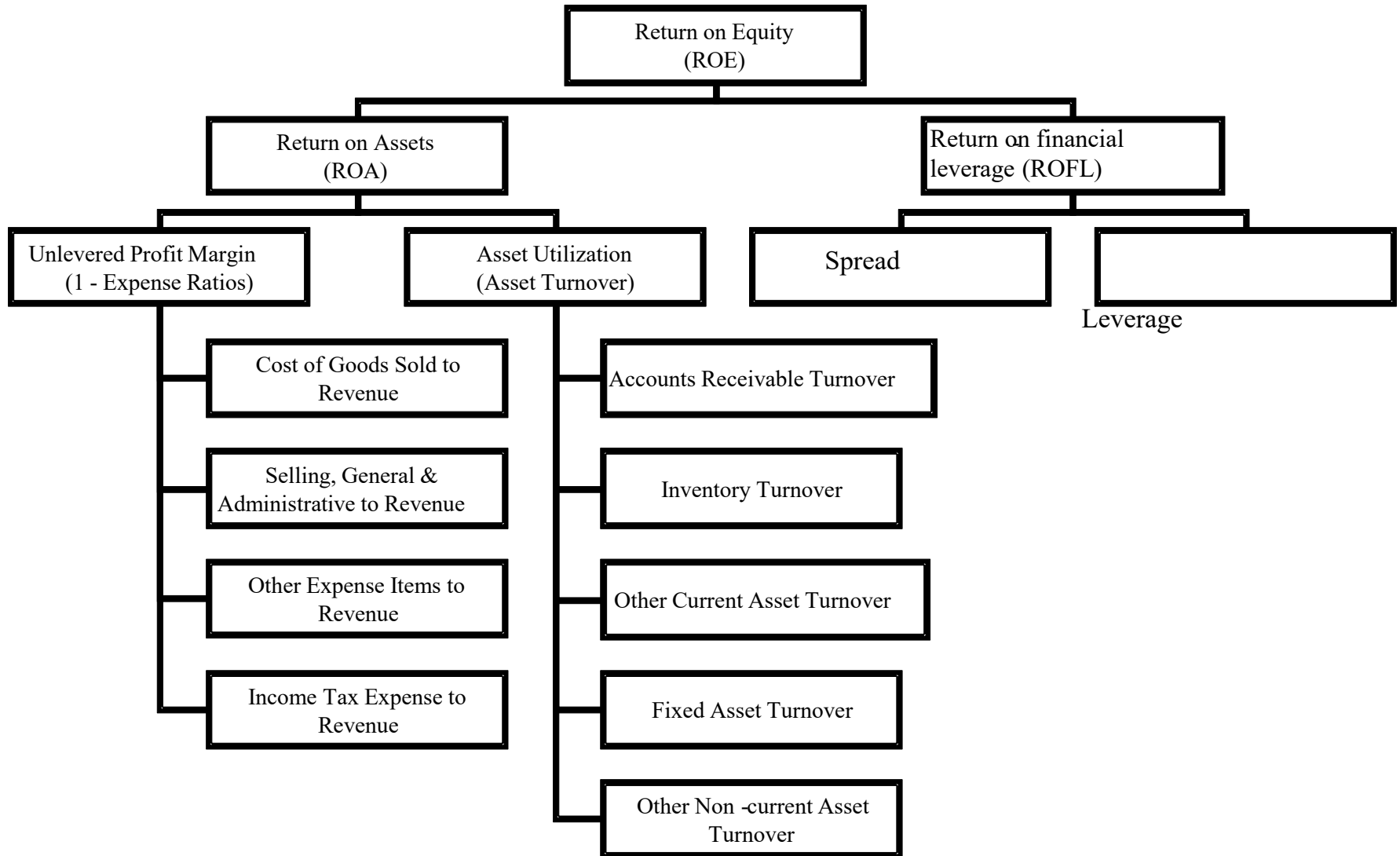
Excess cost of debt over ROA caused company B's ROE to fall below the ROA by 0.8%.

Deriving the relationship between ROE, ROA, and ROFL

$$\begin{aligned}
 ROE &= \frac{\text{Net Income}}{\text{Average Equity}} \\
 &= \frac{NI + I(1 - t) - I(1 - t)}{\text{Average Equity}} \\
 &= \frac{NI + I(1 - t)}{\text{Average Equity}} - \frac{I(1 - t)}{\text{Average Equity}} \\
 &= \underbrace{\left(\frac{NI + I(1 - t)}{\text{Average Assets}} \right)}_{ROA} \times \left(\frac{\text{Average Assets}}{\text{Average Equity}} \right) - \underbrace{\left(\frac{I(1 - t)}{\text{Average Debt}} \right)}_{\text{after tax cost of debt, } R_D} \times \left(\frac{\text{Average Debt}}{\text{Average Equity}} \right) \\
 &= ROA \left(1 + \frac{\text{Average Debt}}{\text{Average Equity}} \right) - R_D \times \left(\frac{\text{Average Debt}}{\text{Average Equity}} \right) \\
 &= ROA + \underbrace{\underbrace{(ROA - R_D)}_{\text{spread}} \times \underbrace{\left(\frac{\text{Average Debt}}{\text{Average Equity}} \right)}_{\text{leverage}}}_{ROFL} \text{ so that} \\
 \Rightarrow ROFL &\equiv \underbrace{\underbrace{(ROA - R_D)}_{\text{spread}} \times \underbrace{\left(\frac{\text{Average Debt}}{\text{Average Equity}} \right)}_{\text{leverage}}}_{\text{}} > 0 \iff ROA - R_D > 0
 \end{aligned}$$

High ROE implies a combination of risky assets (to keep spreads positive) and high leverage. Stated differently, in a highly leveraged business, keeping spreads positive is key to profitability and solvency!

ROA vs. ROE



Liquidity and Solvency Analyses

- Companies can effectively use debt to increase ROE via nonoperating return
- Why don't companies use the maximum debt possible?
- The short answer is that lenders charge higher interest rates for higher debt levels to capture the higher risk creditors face.
- At some point, the cost of additional debt exceeds ROA and additional debt does not make economic sense
- To figure out much debt can a company reasonably manage, liquidity and solvency metrics are useful.

Liquidity and Solvency Analyses

Liquidity Analysis

The analysis of available cash

- Ratios used to assess the degree of liquidity
 - Current ratio
 - Quick ratio
 - Operating cash flow to current liabilities

Solvency Analysis

The ability to generate sufficient cash in the future

- Ratios used to assess the degree of solvency
 - Debt-to-equity
 - Times interest earned

Current Ratio

- Reflects a company's ability to meet its current liabilities by liquidating its current assets.

Current Assets

Those assets that the company expects to convert into cash within the next accounting cycle

Current Liabilities

Those liabilities that come due within the next year

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

Quick Ratio

- Reflects a company's ability to meet its current liabilities without liquidating inventories that could include markdowns
 - Excludes inventories and prepaid assets so a more conservative measure of liquidity.

Quick ratio =

Cash + Short-term securities + Accounts receivable

Current liabilities

Operating Cash Flow to Current Liabilities

- Relates the net amount of cash from operating activities to the amount of current payment obligations
- A key factor in the ultimate ability of a company to pay its debts

$$\text{Operating Cash Flow to Current Liabilities} = \frac{\text{Cash flow from operations}}{\text{Average current liabilities}}$$

Solvency Analysis

- Solvency refers to a company's ability to meet its debt obligations over the longer-term
- Two general approaches to measuring solvency:
 1. Use balance sheet data to assess the proportion of capital raised from creditors
 2. Use income statement data and assesses the profit generated relative to debt payment obligations (aka coverage ratios)

Solvency Analysis: Balance Sheet Data

$$\text{Liabilities-to-equity ratio} = \frac{\text{Total liabilities}}{\text{Stockholders' equity}}$$

- Measures how reliant a company is on creditor financing versus equity financing
- A higher ratio indicates less solvency, and more risk
- A variant of this ratio considers a company's long-term debt divided by equity.
 - This approach assumes that current liabilities are repaid from current assets (so-called self-liquidating).
 - Thus, it assumes that creditors and stockholders need only focus on the relative proportion of long-term capital.

Solvency Analysis: Coverage Ratios

- The second type of solvency analysis compares profits to liabilities
- This approach assesses how much operating profit is available to cover debt obligations.
- A common measure for this type of solvency analysis is the *times interest earned ratio*:

$$\text{Times interest earned} = \frac{\text{Earnings before interest and taxes}}{\text{Interest expense, gross}}$$

Solvency Analysis: EBITDA Coverage Ratio

$$\text{EBITDA coverage} = \frac{\text{Earnings before interest and tax (EBIT)} + \text{Depreciation} + \text{Amortization}}{\text{Interest expense, gross}}$$

- Recall that EBITDA is a non-GAAP metric.
- EBITDA coverage ratio measures company's ability to pay interest out of current profits
- EBITDA coverage ratio is more widely used than the times interest earned ratio because depreciation and amortization do not require a cash outflow
- Always higher than times interest earned ratio

Solvency Analysis: Cash Flow Coverage Ratio

- Cash from operations to total debt measures the ability to generate additional cash to cover debt payments as they come due

$$\text{Cash from operations to total debt} = \frac{\text{Cash from operations}}{\text{Short-term debt} + \text{Long-term debt}}$$

- Free operating cash flow to total debt considers excess operating cash flow after cash is spent on capital expenditures

$$\text{Free operating cash flow to total debt} = \frac{\text{Cash from operations} - \text{CAPEX}}{\text{Short-term debt} + \text{Long-term debt}}$$

Credit Ratings

- A credit rating is an *opinion of an entity's creditworthiness*, captured in alpha-numeric scales:

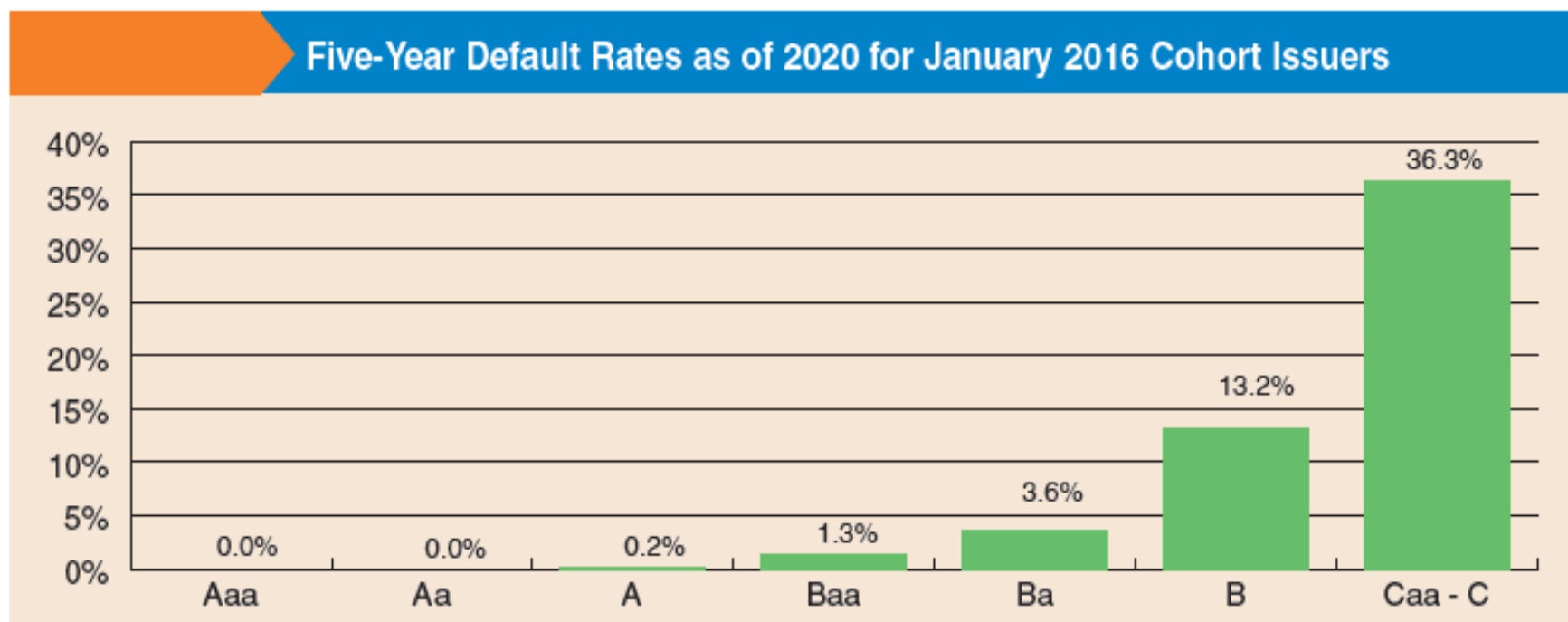
Investment
Grade

Noninvestment
Grade

S&P	Moody's	Fitch	Description
AAA	Aaa	AAA	Prime Maximum Safety
AA+	Aa1	AA+	High Grade, High Quality
AA	Aa2	AA	
AA-	Aa3	AA-	
A+	A1	A+	Upper-Medium Grade
A	A2	A	
A-	A3	A-	
BBB+	Baa1	BBB+	Lower-Medium Grade
BBB	Baa2	BBB	
BBB-	Baa3	BBB-	
BB+	Ba1	BB+	Non-Investment Grade
BB	Ba2	BB	Speculative
BB-	Ba3	BB-	
B+	B1	B+	Highly Speculative
B	B2	B	
B-	B3	B-	
CCC+	Caa1	CCC	Substantial Risk In Poor Standing
CCC	Caa2		
CCC-	Caa3		
CC	Ca		Extremely Speculative
C	C		May be in Default
		DDD	Default
		DD	
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Credit ratings

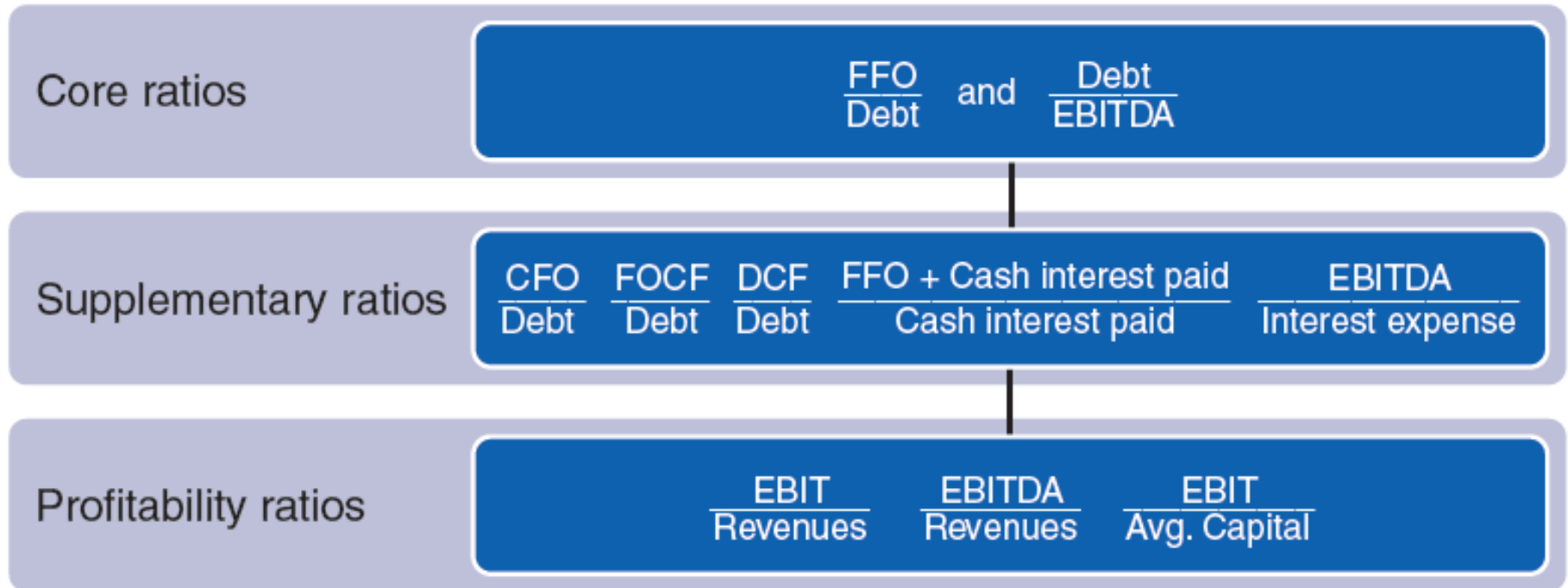
- On average credit ratings predict loan default
- Default rates increase as Moody's credit ratings weaken



How are credit ratings determined?

- Each credit rating agency has its own *unique approach* to credit rating
- Rating agencies have access to information not available to other lenders
- Models are proprietary but all include at least three types of inputs:
 - Macroeconomic statistics
 - Industry data,
 - Company specific information

GAAP and Non GAAP metrics play a crucial role in determining ratings...



- Funds from operations (FFO): EBITDA, minus cash interest paid minus cash tax paid.
- Free operating cash flow (FOCF): CFO minus capital expenditures.
- Discretionary cash flow (DCF): Free operating cash flow (FOCF) minus cash dividends paid on common and preferred stock, less share buybacks.

GAAP and Non GAAP metrics play a crucial role in determining financial risks..

Ratio Values for Different Risk Classes of Corporate Debt*										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Debt/ EBITDA	EBITA / Interest Expense	Revenue (in \$ billion)	EBITA Margin	Operating Margin	FFO ¹ / Debt	(FFO + Interest Expense)/ Interest Expense	EBITA/ Average Assets	CAPEX/ Depreciation	Revenue Volatility
Aaa.....	1.1	63.0	146.0	38.26%	34.1%	83.5%	62.4	21.0%	1.6	15.5
Aa.....	1.3	29.7	143.2	22.85%	22.2%	65.4%	30.9	12.0%	1.2	15.6
A.....	2.1	14.4	22.4	19.71%	16.9%	41.7%	16.1	15.1%	1.2	12.7
Baa.....	2.6	8.4	10.7	17.04%	15.0%	32.6%	10.4	10.2%	1.2	12.2
Ba.....	3.3	5.1	3.5	14.32%	12.3%	24.9%	6.7	10.8%	1.1	15.8
B.....	6.0	2.1	0.9	12.28%	7.4%	9.7%	2.7	7.2%	1.1	18.8
Caa-C....	9.5	0.8	0.5	7.08%	2.2%	3.1%	1.4	4.4%	0.8	15.8

* Table reports 2022 median values: from *Moody's Financial Metrics™ Key Ratios by Rating and Industry for North American Nonfinancial Corporates: 2022 Update*, November 2, 2022.

- Debt is increasingly risky as we move from the first row to the last
- For the most part, ratios weaken from row to row